STAT 527
Multiple Hypothesis Testing

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References:

- Efron, Hastie, "Computer Age Statistical Inference"
- Benjamini and Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," 1995.

Summary:

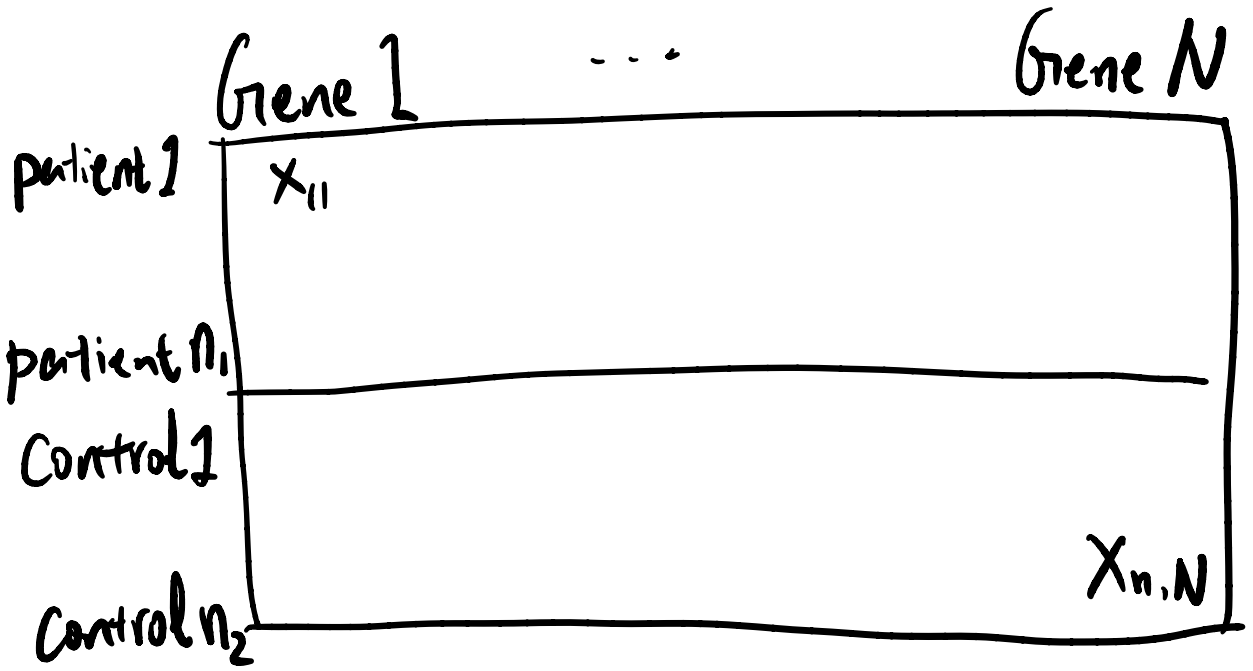
The goal is to perform multiple statistical hypothesis tests simultaneously. Two common approaches for extending the notion of the Type I error to multiple testing include:

- Familywise Error Rate (FWER), which can be controlled by the Bonferroni correction or Holm's procedure, when the p-values (with arbitrary dependence) are available.
- False Discovery Rate (FDR), which can be controlled by the Benjamini-Hochberg method when independent p-values are available, or the knockoff filter in

more general settings.

Example 1: two-sample t-test

Testing if gene expressions are related to having a disease (see figure).



Assume:

- $x_{ij} \sim \mathcal{N}(\beta_{1j}, \sigma^2)$ patient $1 \leq i \leq n$;
- $x_{ij} \sim \mathcal{N}(\beta_{0j}, \sigma^2)$ control $n_1 + 1 \leq i \leq n_1 + n_2 = n$;

Want to test:

$H_{0j} \colon \beta_{1j} = \beta_{0j}$, vs $H_{1j} \colon \beta_{1j} \neq \beta_{0j}$

Define

$$H_0 := \{j \colon \text{ s.t. } H_{0j} \text{ is true}\}$$

Method:

compute

$$t_j = \frac{\frac{1}{n_1}\sum_{1\leq i\leq n_1} x_{ij} - \frac{1}{n_2}\sum_{n_1+1\leq i\leq n} x_{ij}}{\hat{\sigma}\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}} \tag{1}$$

$$\hat{\sigma}^2 := \frac{\sum_{1\leq i\leq n_1}(x_{ij}-avg_{1j})^2 + \sum_{n_1+1\leq i\leq n}(x_{ij}-avg_{2j})^2}{n-2} \tag{2}$$

where

$$avg_{1j} := \frac{1}{n_1}\sum_{1\leq i\leq n_1} x_{ij}; \tag{3}$$

$$avg_{2j} := \frac{1}{n_2}\sum_{n_1+1\leq i\leq n} x_{ij}; \tag{4}$$

therefore t_j follows t-distribution with $dof = n - 2$.

(why not define $\hat{\sigma}^2 := \frac{1}{n-1}\sum_{1\le i\le n}(x_{ij} - avg_j)^2$, where $avg_j := \frac{1}{n}\sum_{1\le i\le n} x_{ij}$?)

p-values

Using t_j's and the cdf of the t-distribution, we obtain p_j, $j = 1, \ldots, N$.

$p_j = \mathbb{P}[|T| \ge |t_j|]$
$T \sim$ t-distribution

Example 2: linear regression

Let $X \in \mathbb{R}^{n\times N}$ be a deterministic design matrix, $n \ge N$, $Y = X\beta + \epsilon$, and ϵ is white Gaussian.

Want to test:

$$H_{0j}\colon \beta_j = 0, \text{ vs } H_{1j}\colon \beta_j \neq 0$$

p-values

Let $t_j := \frac{\hat{\beta}_j^{OLS}}{\hat{\sigma}\sqrt{[(X^\top X)^{-1}]_{jj}}}$. Use the cdf of the t-distribution to obtain p_j, $j = 1, \ldots, N$.

$\mathbb{P}[|t_j| \geq a \,|\, |t_j| \geq b]$ ↑ as b ↑

Test a single hypothesis with confidence α:

Reject H_{0j} iff $p_j \leq \alpha$.

Family-wise error rate (FWER)

Reject H_{0j} if $p_j < \frac{\alpha}{N}$.

$$FWER = \mathbb{P}\left[\bigcup_{j \in H_0} \left\{p_j \leq \frac{\alpha}{N}\right\}\right] \leq \frac{|H_0|\alpha}{N} \leq \alpha.$$

$\leq \mathbb{P}[\bigcup_{j \in H_0} \{\text{reject } H_j\}]$

union bound

Bonferroni correction

$p_j \sim \text{unif}[0,1]$ if $j \in H_0$

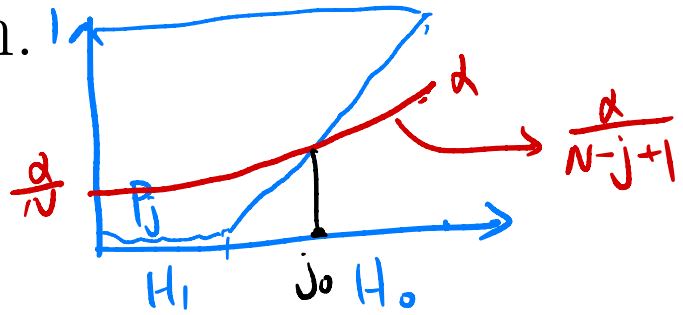
Holm's procedure

Order p-values as $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(N)}$. Let

$$j_0 := \min\{j : p_{(j)} > \frac{\alpha}{N - j + 1}\}. \tag{5}$$

Reject $H_{0(j)}$ if $j < j_0$. Regardless of the dependency of the p-values, it can be shown that Holm's procedure

controls FWER at α, while being uniformly more powerful than the Bonferroni correction.



Idea of Holm's procedure

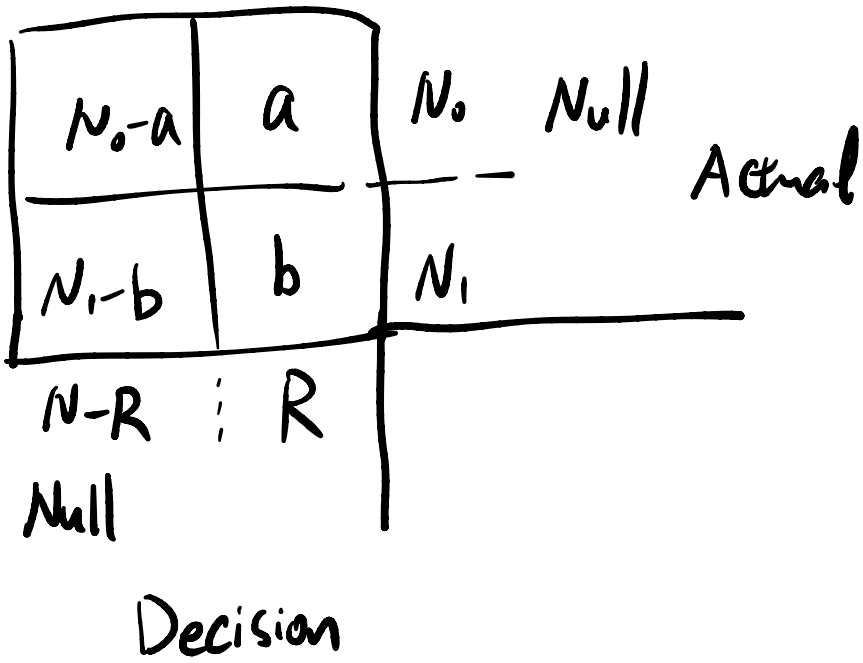
Holm's procedure controls the FWER at α, because of the following claim: If we wrongly reject some true hypothesis, there is a true hypothesis whose p-value is at most $\frac{\alpha}{|H_0|}$.

The procedure is uniformly more powerful than the Bonferroni correction because Holm's procedure will fail to reject a hypothesis only if the Bonferroni correction fails too. Indeed, Holm's procedure will become the Bonferroni correction if the $\frac{\alpha}{N-j+1}$ threshold is de-

creased to $\frac{\alpha}{N}$.

False discovery rate

$FDR := \mathbb{E}[\frac{a}{R}]$ (see figure)

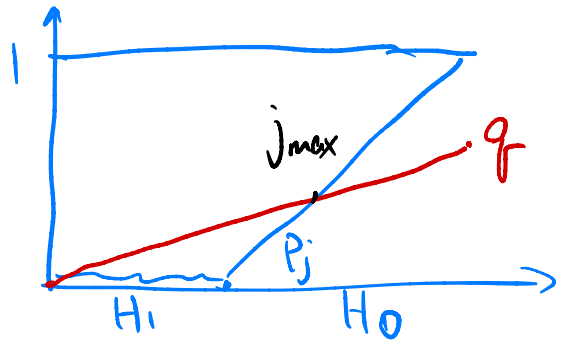


Goal: $FDR \leq q$.

Benjamini-Hochberg

$$j_{max} := \max \left\{ j : p_{(j)} \leq \frac{jq}{N} \right\}; \tag{6}$$

reject $H_{0(j)}$ if $j \leq j_{max}$.



Key result for BH

If p-values are independent as in Example 1 (or more generally, positively correlated as in Example 2), then BH guarantees

$$FDR \leq \frac{N_0}{N} q \leq q. \tag{7}$$

Proof: see Benjamini-Yekutieli 2001, "The control of the false discovery rate in multiple testing under dependency."